DelvMap: Completing Residential Roads in Maps Based on Couriers' Trajectories and Satellite Imagery

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***Abstract*— The updated residential-level fine-grained digital map is essential for last-mile delivery. However, many of those low-level roads are not recorded in maps due to the high mapping costs. With the digitization of the logistics industry, couriers' trajectories become a promising data source to complete missing roads in maps. Existing trajectory-based map updating work rely on heavy parameter tuning to overcome the positioning error issue due to their unsupervised nature, and are not able to handle issues of unreliable road indicators and skewed data distributions. To tackle those challenges, in this article, we propose a framework DelvMap to complete missing roads in maps based on couriers' trajectories and satellite images. DelvMap first leverages a dual signal fusion network (DSFNet) to extract an inferred map from both data modalities, which fully exploits the positive and negative signals in the satellite images to fuse with roads indicated from trajectories, then employs a map completion algorithm to complete the existing map with the inferred map, the connection strategy of which is adaptive to the number of traversing trajectories. Experiments show DelvMap outperforms baselines by at least 11.0% in TOPO F1 on the real-world dataset. Finally, we demonstrate a multimodal map updating system based on DelvMap.**

***Index Terms*— Delivery data mining, map updating, multimodal learning.**

I. Introduction

THE last-mile delivery is mainly conducted in the residential area, which requires residential-level fine-grained digital maps, especially for fresh couriers or autonomous delivery vehicles. However, as shown in Fig. 1, authorized mapping agencies focus on keeping the map with high road levels updated, while are not responsible for maintaining residential roads, which leaves significant portion of them unrecorded in maps, and thus incurs the efficiency penalty to the location-based applications in the last-mile delivery, e.g., navigation, arrival time prediction [1], and order pickup.

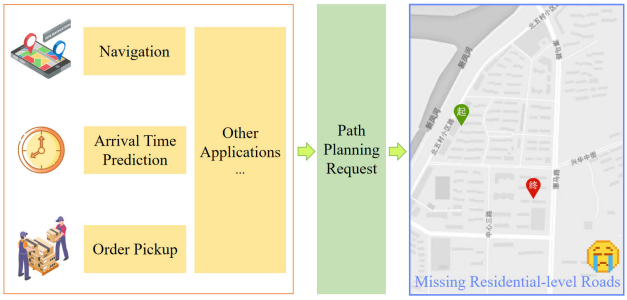


Fig. 1. Background of the map updating.

Fortunately, with the digitization of the logistics industries in recent years, massive trajectories recorded by couriers' PDAs are accumulated, which indicate the existence of roads in the urban space, especially in the residential area. Those data have great potential to complete residential roads in maps. However, it is nontrivial due to the following challenges.

1) *GPS Positioning Errors:* The average positioning errors of GPS devices are about 5 to 15 m, which make the recorded trajectories deviate for some distance from the roads, where the courier is moving on. What's worse, since buildings are densely distributed in the residential area, the positioning errors in such urban canyons are even larger due to multipath effects as the red points shown in Fig. 2(a).
2) *Unreliable Road Indicators:* Unlike cars, couriers frequently enter buildings to deliver parcels, leading to the generation of many GPS points within indoor environments (in case GPS signals are available). The existence of those GPS points, however, does not imply roads as shown in Fig. 2(b).
3) *Highly Imbalanced Data Distributions in Spaces:* The movements of couriers are driven by the parcels they need to deliver. Since the logistics demand follows a long-tailed distribution in space, as shown in Fig. 2(c), the number of times each road traveled by couriers varies greatly, and some roads might never be traveled even if we accumulate the trajectory data for a long time period.

Most existing work termed the problem as map updating, which mainly aimed to complete motorways since most

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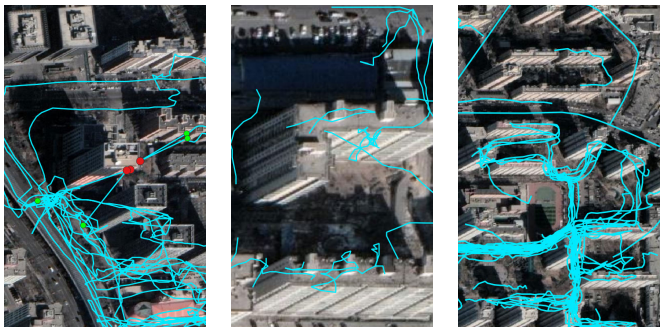
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(a) (b) (c)

Fig. 2. Challenges of the map updating. (a) Positioning errors. (b) Unreliable road indicators. (c) Imbalanced data distributions.

trajectories are from the floating cars [2], [3], [4]. VitalAlley [5] is the first to leverage pedestrians' trajectories to discover the missing residential-level roads. But no matter which data source is used, they all use unsupervised methods to overcome the positioning error issue, e.g., generating new roads by trajectory clustering [3] or kernel density estimation (KDE) [5], which require heavy parameter-tuning efforts and are not scalable to data sizes. In addition, the unreliable indicator issue and the imbalanced data issue are also not well-handled.

Map generation is a similar problem, which generates the map from the scratch. Recently, it is studied by the supervised approach [6], which learns a mapping from trajectories to road centerlines by a neural network, to overcome the positioning errors. To further tackle the issue of the imbalanced data distribution, the multimodality fusion-based map generation, which fuses the satellite imagery [7], [8], [9], [10], leads the trend. However, those work might cause the replacement of the existing roads, which cannot be directly adopted to tackle our problem, since the map updating usually reserves the existing map, and only connects newly discovered missing roads to it. Because the existing map should be much more reliable than the inferred one given that it is manually created by humans, e.g., authorities or web volunteers, and may contain more semantic-rich road attributes, e.g., the road name. Apart from that, the issue of unreliable road indicators is not explicitly handled in these work, which is not rare in our scenario, since the moving constraints of couriers are much loose than floating cars on motorways.

To this end, in this article, we propose a novel map updating framework, i.e., delivery map (DelvMap in short), to complete the missing residential roads in maps based on couriers' trajectories and satellite images. DelvMap first generates an inferred map based on couriers' trajectories and satellite images via a dual signal fusion network (DSFNet) to overcome issues of the positioning errors and imbalanced data distribution of trajectories. In addition to extracting and fusing roads from the satellite images, DSFNet also extracts buildings from them, which helps denoise the trajectory modality at an early stage and explicitly handles the unreliable road indicator issue. After the map is inferred, the new roads in the inferred map are connected to the existing map, the connection strategy of which is adaptive to the number of traversing trajectories to balance the trade-off between the topology correctness and the data sparsity.

Our contributions can be summarized as follows fourfolds.

1) We make the first attempt to complete residential roads in maps based on both couriers' trajectories and satellite images, and identify the unique challenge of residential road completion, i.e., unreliable road indicators.
2) We present a multimodal map updating framework DelvMap. It is composed of a novel dual signal fusion neural network, i.e., DSFNet, and an adaptive map completion strategy. DSFNet simultaneously fuses both positive and negative signals from the satellite images via STFuse and BTFuse modules to achieve accurate and high-coverage road extraction. The adaptive map completion strategy considers the trade-off between transition evidence and spatial proximity to achieve better connectivity.
3) Extensive experiments based on real-world couriers' trajectories from JD Logistics as well as satellite images demonstrate the effectiveness of DelvMap, which outperforms baselines by at least 11.0% in TOPO F1.
4) We present a map updating system based on DelvMap, which has been deployed in JD Logistics for DelvMap updating. The map serves as the infrastructure to enable downstream tasks, e.g., the route planning and the package pick-up scheduling.

II. Preliminaries

Definition 1 (Trajectory): A trajectory is a sequence of spatio-temporal points, denoted as $tr =< p_1, p_2, \ldots, p_n >$, where each point $p = (x, y, t)$ indicates the physical presence at a location (x, y) (e.g., longitude and latitude) at time t. Points in a trajectory are organized chronologically.

Definition 2 (Map): A map is an undirected graph with geospatial coordinates, denoted as $Map =< V, E >$, where each vertex $v \in V$ is associated with a location (x, y), and each edge $e \in E$ is a vertex tuple (u, v), denoting a road connecting u and v.

Note that the map can also be modeled as the directed graph. However, in this work, we focus on producing a map that can be used for last-mile delivery, in which scenario the moving direction is usually not restricted for pedestrians and small tricycles.

Problem Definition: Given a set of couriers' trajectories $\mathcal{T} = \{tr_1, tr_2, \ldots, tr_n\}$ and satellite images of the area of interest, complete the map from $\text{Map}_{\text{existing}}$ to $\text{Map}_{\text{complete}}$.

III. Methodology

A. Overview

Existing map updating algorithms [2], [3], [4], [5] follow a three-step framework.

1) Firstly, they match trajectories on the existing map [11], and only reserve the unmatched trajectory data.
2) Secondly, unmatched trajectories are used to generate the inferred road segments.
3) Finally, the inferred road segments are merged with the existing map to realize the map updating either based on the trajectory transition relationship or the spatial proximity.

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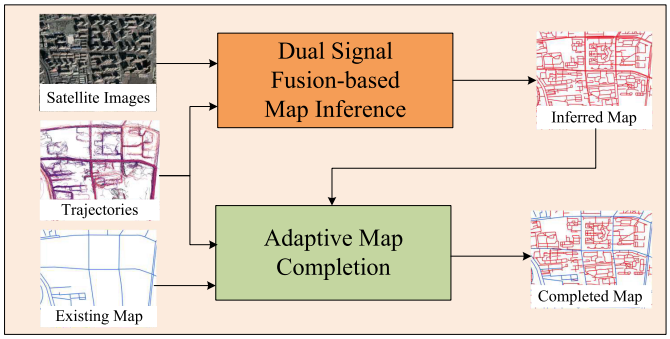


Fig. 3. Framework of DelvMap.

However, such a framework has two limitations, which do not apply to multimodality-based map updating nowadays.

1) The map-matching process itself is very time-consuming, which makes the overall running time of the map updating extremely long, especially when large-scale trajectories are given.
2) Unlike trajectories, it is not straightforward for the satellite image source to only infer the missing roads in the existing map. Though we may mask pixels covered by roads in the existing map, the masked pixels would affect the recognition of roads due to the uncertain neighborhood.

To this end, we present a novel framework for the multimodality-based map updating, i.e., DelvMap, which is presented in Fig. 3. DelvMap first leverages trajectories and satellite images to infer a whole map via *Dual Signal Fusion-based Map Inference* component, and then completes the existing map considering trajectories based on the inferred map via *Adaptive Map Completion* component. In this way, DelvMap avoids the tedious map-matching process and enables accurate pattern recognition from satellite images.

In the following, we explain each component in detail.

B. Dual Signal Fusion-Based Map Inference

In this component, we first propose a convolution neural network DSFNet, to infer the positions of roads from couriers' trajectories and satellite images, and use the inferred roads to construct an inferred map. DSFNet detects areas covered by roads as well as buildings in the satellite images and fuses with the trajectory source in different stages (we name this strategy the hybrid fusion) to generate accurate road predictions.

Similar to most existing work [10], [12], we formulate road detection as an image segmentation task, which performs the segmentation on region tiles. Therefore, we split the region of interest into uniform tiles, detect roads in each tile based on multimodality data, and then concatenate all predicted tiles to extract the inferred map. We explain those steps in detail as follows.

1) Feature Extraction: While obtaining the corresponding satellite image of each region tile is straightforward, the trajectory representation of each region tile should be rendered with additional efforts. Following [6], we render a two-channel tensor from the trajectory source, i.e., the point distribution as well as the line distribution constructed by consecutive points in trajectories. Those distributions are normalized by log scale and blurred by Gaussian to mitigate the imbalanced spatial distribution of trajectories. The point distribution is the most straightforward indicator of the underlying roads, while the line distribution could help with recovering the roads when points are sparse. Some other measurements could also be integrated, e.g., the direction distribution, but we omit them here since those data are rather noisy for couriers' trajectories given that they move more freely compared with floating cars.

2) DSFNet: The design of the road extraction model is inspired by our observations from the data. As we discussed in Section I, the trajectories may not always be reliable indicators of roads, since the couriers can also appear at some locations that are nonroads, e.g., within a residential building. Therefore, if we can detect those nonroad regions where trajectories may exist in advance, we can use this information to denoise the trajectory modality. After that, a much cleaner trajectory modality can be used to fuse with the satellite images to make a more accurate road prediction.

Different from existing multimodal fusion models [7], [8], [9], which only leverage road labels (positive signals) to train the network, we also introduce the nonroad signals (i.e., area covered by buildings) into the model. Therefore, we name the proposed model DSFNet. In the following, we illustrate the structure of DSFNet in detail.

a) Overview: The overview of DSFNet is shown in Fig. 4. It contains three segmentation networks, which serve for predicting the area covered by building $\hat{\mathbf{Y}}_B$ based on the satellite images, roads $\hat{\mathbf{Y}}_R^T$ based on the trajectories, as well as roads $\hat{\mathbf{Y}}_R^S$ based on the satellite images, respectively. During the inference, we only use $\hat{\mathbf{Y}}_R^S$ to extract the inferred map, which fuses all information as we explained in the following. The widely adopted UNet [13] is employed as the backbone. It consists of an encoder which extracts features and downsamples the original input several times, and a decoder, which upsamples the encoded information for several times to recover the original shape of the input. The encoder and decoder have linked shortcuts at corresponding layers.

There are two types of information fusion between two modalities in DSFNet.

1) Building-trajectory fusion (BTFuse) takes the feature map $\mathbf{F}_B$ before the prediction layer for the building segmentation, and fuses it with the trajectory modality just before the first time of pooling, i.e., $\mathbf{F}_T^{(1)}$. It can be regarded as a kind of early fusion.
2) Satellite-trajectory fusion (STFuse) fuses the bottleneck feature maps of trajectory modality to the satellite image modality. It can be regarded as a kind of deep fusion. We explain the implementation details of each module as follows.

b) BTFuse module: BTFuse is designed to denoise the trajectory modality. The pixels with higher values in $\mathbf{F}_B$ should be given stronger negative signals in the trajectory modality. Therefore, the building information-fused trajectory feature map $\widetilde{\mathbf{F}}_T^{(1)}$ is achieved through producting $\mathbf{F}_T^{(1)}$ with a reversed normalized feature maps of $\mathbf{F}_B$

$$\widetilde{\mathbf{F}}_T^{(1)} = \mathbf{F}_T^{(1)} \odot \left(\mathbf{1} - \frac{\mathbf{F}_B}{\max(\mathbf{F}_B)}\right) \tag{1}$$

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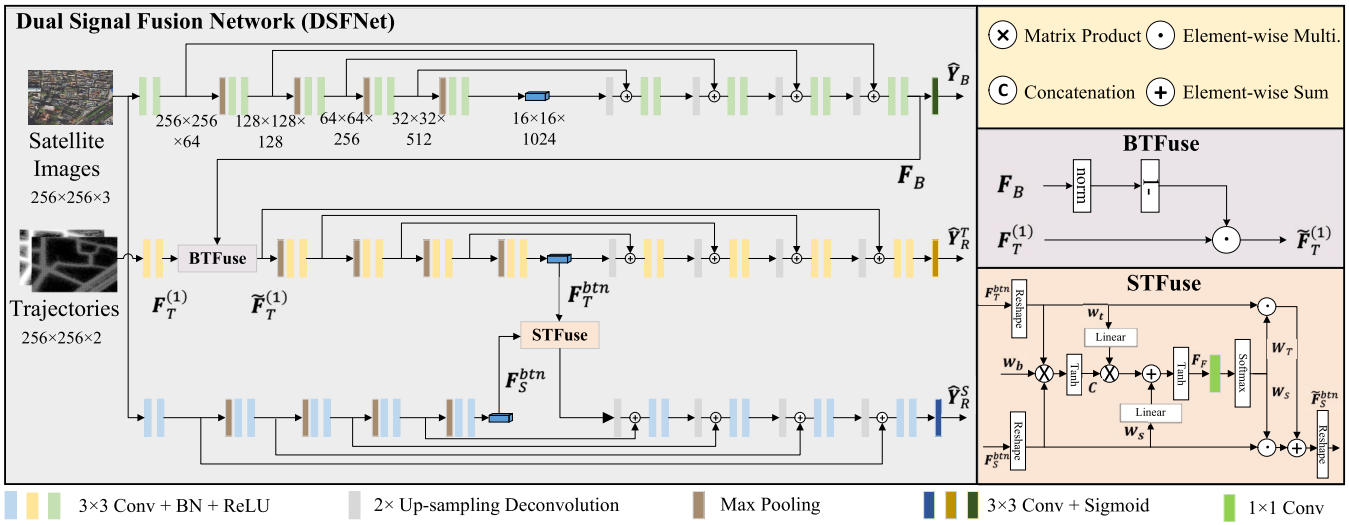


Fig. 4. Structure of DSFNet.

where $\odot$ is the Hadamard product of two tensors, $\max(\cdot)$ is the channel-wise maximum function.

c) STFuse module: STFuse is designed to fuse signals for road detection from the trajectory modality to the satellite image modality, which is conducted in the bottleneck layer of two road segmentation networks. Existing deep-fusion work fuses information based on the convolutional layers [8], [9], which can only capture dependencies in the neighborhood. Co-attention mechanism has been studied in visual and language tasks, such as visual question answering [14], [15], [16], [17] and visual dialog [18]. It offers the potential to better understand the correlations between different modalities and their corresponding features within their feature embedding space. Inspired by this, we devise a co-attention-based fusion method, so that the pairwise dependencies for all pixels in the feature map can be leveraged. STFuse is conducted in the bottleneck layer of two road segmentation networks. It first calculates the pixel-wise dependencies bottleneck features maps from two modalities, which enables us to determine the spatial correspondence between feature maps from distinct images. Then, a weight matrix is introduced to regulate the allocation of information between the two modalities during the fusion process. We detail the fusion process as follows.

To decide how much information in $\mathbf{F}_T^{btn}$ and $\mathbf{F}_S^{btn}$ should be leveraged during the fusion, we aim to learn the pixel-wise weights $\mathbf{W} \in \mathbb{R}^{H_{btn}W_{btn}\times 2}$ for two modalities.

Specifically, we first calculate the pixel-wise similarity between $\mathbf{F}_T^{btn}$ and $\mathbf{F}_S^{btn}$ by reshaping both of them into the shape of $C_{btn} \times (H_{btn}W_{btn})$ and perform the matrix–matrix product similar to the co-attention mechanism [16]. We also introduce learnable parameters $\mathbf{W}_b \in \mathbb{R}^{C_{btn}\times C_{btn}}$ to get the final similarity matrix $\mathbf{C} \in \mathbb{R}^{(H_{btn}W_{btn})\times(H_{btn}W_{btn})}$. In this way, the pairwise dependencies among pixels of two modalities can be captured as shown in the following equation:

$$\mathbf{C} = \tanh(\mathrm{RE}(\mathbf{F}_T^{btn})^\top \mathbf{W}_b \mathrm{RE}(\mathbf{F}_S^{btn})) \tag{2}$$

where $\mathrm{RE}(\cdot)$ is the reshape operation.

Then we obtain the fused feature map $\mathbf{F}_F \in \mathbb{R}^{(H_{btn}W_{btn})\times(H_{btn}W_{btn})}$ based on feature maps of both modalities and the similarity matrix

$$\mathbf{F}_F = \tanh(\mathbf{w}_s \mathrm{RE}(\mathbf{F}_S^{btn}) + (\mathbf{w}_t \mathrm{RE}(\mathbf{F}_T^{btn}))\mathbf{C}) \tag{3}$$

where $\mathbf{w}_s, \mathbf{w}_t \in \mathbb{R}^{(H_{btn}W_{btn})\times C_{btn}}$ are learnable parameters.

After that, a 1×1 convolutional layer is applied to reduce the number of channels to 2. Finally, a softmax layer is applied to obtain $\mathbf{W}$ as shown in the following equation:

$$\mathbf{W} = \mathrm{softmax}(\mathrm{Conv}_{1\times 1}(\mathbf{F}_F)). \tag{4}$$

Based on the generated weights, the final modality-fused bottleneck feature map $\widetilde{\mathbf{F}}_S^{btn} \in \mathbb{R}^{H_{btn}\times W_{btn}\times C_{btn}}$ is obtained by summing up reshaped feature maps from two modalities by weights and recovering the original shape as shown in the following equation:

$$\widetilde{\mathbf{F}}_S^{btn} = \mathrm{RE}(\mathrm{RE}(\mathbf{F}_S^{btn}) \odot \mathbf{W}_S + \mathrm{RE}(\mathbf{F}_T^{btn}) \odot \mathbf{W}_T) \tag{5}$$

where $\mathbf{W}_S = \mathbf{W}[:, 0]$, $\mathbf{W}_F = \mathbf{W}[:, 1]$.

3) Optimization: Since DSFNet involves the inference of the road segments as well as the building contours, the optimization of the model is formulated as a typical multitask learning problem and optimized by the gradient descent. We use the binary cross-entropy (BCE) loss for the building contours segmentation, and the hybrid loss of BCE and Dice [19] for the road segmentation for two modalities, which overcomes the extremely imbalanced label issue given that the roads only cover limited pixels of a region tile. The loss function is formally defined as follows:

$$\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_R(\hat{\mathbf{Y}}_R^T, \mathbf{Y}_R) + \lambda_1 \mathcal{L}_R(\hat{\mathbf{Y}}_R^S, \mathbf{Y}_R) + \lambda_2 \mathcal{L}_{BCE}(\hat{\mathbf{Y}}_B, \mathbf{Y}_B) \tag{6}$$

where $\mathbf{Y}_R$ and $\mathbf{Y}_B$ are the binary ground-truths of roads and building contours, respectively, λ_1 and λ_2 are hyperparameters to balance different learning tasks, and $\mathcal{L}_R$ is the hybrid loss function with a hyperparameter ω: $\mathcal{L}_R(\hat{\mathbf{Y}}, \mathbf{Y}) = \mathcal{L}_{BCE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \omega \mathcal{L}_{\mathrm{Dice}}(\hat{\mathbf{Y}}, \mathbf{Y})$.

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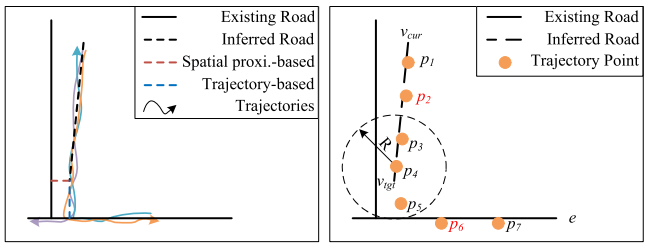


Fig. 5. Illustration of the adaptive map completion. (a) Bad case for the spatial proximity-based connection. (b) Calculation of the transitions.

Note that the ground-truth map represented by the binary matrix only indicates the presence of roads, which may have difficulties in handling cases of road stacking. Though it can be handled by the tensor-represented ground-truth, e.g., Sat2Graph [20], we choose the simpler matrix-represented one since most of roads that need to be attached are in the residential area, where the stacking cases rarely exist.

4) Graph Extraction: After we get the predictions from $\hat{\mathbf{Y}}_R^S$, the graph extraction is conducted, which extracts the maps from the predicted region tiles. We first concatenate all predicted region tiles by DSFNet according to their actual geographical positions, then apply a topological thinning algorithm [21] to make each road one-pixel wide while reserving their topology in the image, and finally adopt a combustion technique [22] to construct the inferred map from the image-represented thinned roads.

C. Adaptive Map Completion

In this component, we fuse the inferred map with the existing map to produce the completed map based on trajectories and the spatial proximity.

1) Intuition: As mentioned in Section III-A, existing map updating work has two road connection strategies.

1) Trajectory-based connection [2], [3], [5], which leverages trajectories to find the topological relationship between the inferred roads and the existing roads as the blue dashed line shown in Fig. 5(a).
2) Spatial proximity-based connection [4], [23], which connects the inferred roads with existing roads based on a distance threshold as the red dashed line shown in Fig. 5(a). Though the former strategy can produce a more accurate topology of the map, it is only applicable to the trajectory-driven map updating problem. On the contrary, the latter strategy can be applied to both satellite image-based and trajectory-based map updating, but the constructed topology might be spurious. Since the roads inferred by us could come from different modalities, we aim to combine the best of both worlds. For inferred road segments associated with sufficient traversing trajectories, we leverage the trajectory-based connection for the accuracy consideration, while for inferred road segments associated with few trajectories or without trajectories, we employ the spatial proximity-based connection for the generalization consideration.

Algorithm 1 Adaptive Map Completion Algorithm

Input: The existing map $Map_{existing}$; the inferred map $Map_{inferred}$; trajectories $\mathcal{T}$, the new vertex threshold TH_{new}, the vertex merging threshold TH_{vm}, the transition support S_{min}, the checking radius R, the trajectory match threshold TH_m.
Output: The completed map $Map_{complete}$.

1: $RN_{complete} \leftarrow copy(RN_{existing})$; $\mathcal{M} \leftarrow HashMap()$;
2: **for** $v \in RN_{inferred}.V$ **do**
3: **if** $v \in \mathcal{M}$ **then**
4: **continue**;
5: **if** $range_query(v, RN_{existing}, TH_{new})$ is not empty **then**
6: **continue**;
/*Start examing at an unvisited missing vertex.*/
7: $\mathcal{M}[v] \leftarrow v$;
8: $q \leftarrow Queue()$; $q.add(v)$;
9: **while** q is not empty **do**
10: $v_{cur} \leftarrow q.pop_last()$;
/*Check each road connected to v_{cur}.*/
11: **for** $v_{tgt} \in adjacent_vertices(RN_{inferred}, v_{cur})$ **do**
12: $E_{nbhd} \leftarrow range_query(v_{tgt}, RN_{existing}, TH_{vm})$;
/*No nearby existing roads, directly add the new road.*/
13: **if** E_{nbhd} is empty **then**
14: $\mathcal{M}[v_{tgt}] \leftarrow v_{tgt}$;
15: $RN_{complete}.add_edge(\mathcal{M}[v_{cur}], \mathcal{M}[v_{tgt}])$;
16: $q.add(v_{tgt})$;
17: **continue**;
/*Otherwise, connect and add the new road.*/
18: $e_{tmp}, cnt \leftarrow get_max_transit_edge(E_{nbhd}, v_{cur}, v_{tgt},$
19: $\mathcal{T}, R, TH_m)$;
20: **if** $cnt \geq S_{min}$ **then**
/*Trajectory-based connection.*/
21: $e_{conn} \leftarrow e_{tmp}$;
22: **else**
/*Spatial proximity-based connection.*/
23: $e_{conn} \leftarrow get_min_distance_edge(E_{nbhd}, v_{tgt})$;
24: $v_{proj} \leftarrow calculate_projection(v_{tgt}, e_{conn})$;
25: $\mathcal{M}[v_{tgt}] \leftarrow v_{proj}$;
26: $RN_{complete}.split_edge(e_{conn}, v_{proj})$;
27: $RN_{complete}.add_edge(\mathcal{M}[v_{cur}], \mathcal{M}[v_{tgt}])$;
28: **return** $RN_{complete}$;

2) Main Idea: The main idea of the adaptive map completion is that for each road in the inferred map, if it is far away from the existing map, we directly add it to the existing map; if one of its vertices is close to the existing map, we merge it to the existing map based on the trajectory or the spatial proximity, which is adaptive to the number of trajectories affiliated with the inferred road and the existing road to connect, and then add the inferred road.

3) Implementation: The pseudocode is given in Algorithm 1. We first use a hashmap $\mathcal{M}$ to maintain the mapping from vertices in the inferred map to the vertices after merging with the existing map (Line 1). The iteration process is similar to [23]. For each vertex in the inferred map, we check whether it is far away from the existing map based on a distance threshold $\mathrm{TH}_{\mathrm{new}}$ (Line 5). The vertex met the criteria is highly likely to be a missing vertex in the existing map. Once such a vertex is found, we conduct a depth-first search to check all inferred roads in the connected component to determine whether they should be added to complete the map (Lines 8–27). For each road in the

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connected component, if it is not close to the existing map (quantified by TH_{vm}), we directly add the inferred road to the existing map (Lines 13–17). Otherwise, we try to merge the closest vertex of the inferred road to the existing map based on the trajectory or the spatial proximity depending on the number of trajectories affiliated with the inferred road and the existing road to connect, and then add the inferred road (Lines 19–27). Since the topology constructed by the trajectories are more reliable, we first try to leverage trajectories to merge the vertex, and if it failed due to insufficient transitions (quantified by the transition support $S_{\min}$), we then use the spatial proximity-based connection (Lines 19–23).

One of the core function is get_max_transit_edge, which takes the candidates of existing roads to be connected E_{nbhd}, inferred road $(v_{\mathrm{cur}}, v_{tgt})$, the trajectories $\mathcal{T}$, checking radius R, as well as a trajectory match threshold TH_m as input, and gives the existing road with the maximum number of transitions from or to the inferred road e_{tmp} as well as the number of transitions cnt. We first filter all trajectories which have at least one point within TH_{vm} to v_{tgt} by a range query. Fig. 5(b) show one such trajectory. For the nearest trajectory point to v_{tgt}, i.e., p_4, we find its precursor and the successor that is just R away from p_4, i.e., p_2 and p_6. If such two points just correspond to the existing road and the inferred road (the perpendicular distance from the point to the road is less than TH_m), the number of transitions between the pair of roads is accumulated. Introducing R helps to filter some spatially close trajectory points due to the high sampling rate or the stay event, and makes the transition between roads more obvious.

D. Complexity Analysis

1) Dual Signal Fusion-Based Map Inference: Referring to [24], the computational cost of a single CNN layer is mainly dominated by matrix products. The map inference module DSFNet uses three parallel U-net structures, resulting in an overall complexity of $\mathcal{O}(3\mathrm{NCd}^2)$, where N, C, d represents the number of samples, the number of CNN layers and the dimension of feature vectors, respectively. Additionally, any pooling operation applied to input data does not exceed linear time. Therefore, the overall runtime remains linear to N.

2) Adaptive Map Completion: Though the algorithm is iterated over the vertex of the inferred map, it essentially checks whether each inferred road should be added to the existing map. For each inferred road, it checks its spatial closeness to roads in the existing map. And during the checking, the transition relation is calculated via trajectories. Since the associated trajectories can be efficiently retrieved via the spatial index, thus, the overall time complexity of the algorithm is $\mathcal{O}(|E_{\mathrm{inferred}}||E_{\mathrm{existing}}|\log|\mathcal{T}|)$. In reality, there are only limited roads in the existing map that are close to a road in the inferred map.

The complexity of the overall process of map completion is composed of the above two parts. The inference time of deep learning models is linear and much smaller than that of unsupervised methods. The completion strategy only handles a limited number of roads. Therefore, the framework is practical.

TABLE I
DATA DESCRIPTIONS

	Train & Val	Test
Region Size (km^2)	8.25	4.06
# of Couriers	190	61
# of Trajectories	143,455	28,065
# of Points	36,067,237	8,514,534
Existing Map (km)	284.84	124.25
Ground Truth Map (km)	533.72	226.43

IV. EXPERIMENTS

A. Datasets

Our datasets consist of the existing maps, couriers' trajectories, satellite imagery as well as the ground-truth completed maps and building contours, which are all obtained from four delivery regions with a total coverage area of 12.31 km^2 in Beijing and detailed as follows.

Existing Maps: The existing maps are downloaded from OpenStreetMap (OSM),[1] which are contributed by web volunteers but lack many low-level roads, especially in the residential area.

Couriers' Trajectories: The couriers' trajectories are collected from JD Logistics over 20 months (from January 1, 2018 to September 1, 2019). Several preprocessing algorithms, e.g., noise filtering, stay point removal, and trajectory segmentation are applied. The average sampling rate of the trajectories is 13.51 s.

Satellite Imagery: The satellite imagery is crawled from AMap.[2]

Completed Maps and Building Contours: The completed map and building contours are labeled by five students who are familiar with the locality using QGIS.[3]

We perform the training and validation on datasets of three delivery regions and evaluate/test in the remaining delivery region. The image representations of the existing maps and ground-truth maps are generated by masking two cells surrounding the road centerlines. For methods that require partitioning regions into tiles, the resolution of each region tile is 256×256, with each pixel covering 1×1 m in consistent with existing multimodal road extraction work [8], [9]. The dataset contains 750 disjoint tiles consisting of satellite images and trajectory data. The tiles for training, validation, and testing are randomly split with a ratio of 8:1:1. All training samples are rotated, flipped, and blurred for data augmentation during the training. The detailed statistics of the datasets are given in Table I.

B. Experimental Settings

1) Baselines: We compare DelvMap with five representative trajectory-driven map updating baselines as well as two satellite image-based map updating baselines as follows.

1) CrowdAtlas [2] is the earliest proposed trajectory-driven map updating method. It first clusters unmatched trajectories by a hierarchical clustering algorithm based on the

[1] https://www.openstreetmap.org/
[2] https://amap.com/
[3] https://www.qgis.org/

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Hausdorff distance, then leverages the polyline fitting to discover the road centerlines, and finally discovers the topology based on the transitions indicated in trajectories between existing roads and the discovered roads.

2) VitalAlley [5] is similar to CrowdAtlas, which is also based on trajectory clustering. The major difference is that it leverages the shortest path search over the density map of trajectory points in a cluster to find the road centerlines, and performs the density clustering on the last matched trajectory points to find the interfaces of the inferred roads and the existing roads.
3) GLUE [3] is also a trajectory-driven map updating method, which is based on trajectory point clustering. It devises a greedy-based linkage clustering algorithm to cluster the unmatched trajectory point based on the distance and the direction similarity, then estimates the centerlines and constructs the topology via a method similar to CrowdAtlas.
4) HyMU [4] is a hybrid trajectory-driven map updating method. It first separately clusters trajectory points as well as segments formed by consecutive trajectory points to generate two types of road candidates, then those two types of candidates are further clustered and fit to generate the road centerlines. Finally, the inferred roads are connected to the existing map based on the spatial proximity.
5) DeepMG [6] +trFuse combines the recently deep learning-based trajectory-driven map generation method DeepMG and the trajectory-based connection strategy to fuse the generated maps with the existing maps. The core component of DeepMG is T2RNet, which is a UNet-based multiview multitask segmentation network.
6) RoadTracer+spFuse [23] is one of few work focusing on map updating based on satellite images. It leverages a universal map fusion method that fuses the inferred map predicted by arbitrary road extraction models to the existing one based on spatial proximity. We rename the fusion method as "spFuse" for clarity. We choose RoadTracer [25] as the road extraction model given its superior performance as reported by [23].
7) SDUNet [26]+spFuse is a novel satellite image-based road extraction method, which fuses the multilevel features and global prior information of road networks by leveraging a hybrid approach that integrates spatial CNN-based segmentation and densely connected blocks. We take SDUNet as the road extraction model, and leverage spFuse to connect with existing maps.

2) Variants: We also compare DelvMap with the following variants to show the effectiveness of different components of DelvMap.

1) DelvMap-nDSFNet, which replaces DSFNet with an existing multimodal map generation network that achieves the best performance in our datasets, i.e., DeepDualMapper [9].
2) DelvMap-spFuse, which only leverages spFuse to fuse the inferred map with the existing map, i.e., only the spatial proximity-based connection is employed.
3) DelvMap-trFuse, which only leverages the transition relationship in trajectories to fuse the inferred map with the existing map, i.e., only the trajectory-based connection is employed.

3) Evaluation Metrics: Existing work to evaluate the map updating only cares about the geometrical accuracy of the inferred road segments, which is determined by human [2] or quantified by some geometrical metrics [3], [4], [5]. However, since the updated map essentially is a geospatial graph, its topological accuracy should also be considered.

In the similar problem, i.e., map generation, there are two de-facto metrics to evaluate the quality of the generated map, i.e., GEO metric and TOPO metric [27]. The former evaluates the geometrical accuracy of roads, which essentially is pixel-wise comparison in Precision, Recall, and F1 between the generated map and the ground-truth map. The latter compares the network-based reachable grid cells of both maps from randomly selected starting locations, calculates Precision, Recall and F1 based on reachable grid cells, and reports the average score over several times of the random sampling.

We adapt GEO metric and TOPO metric to evaluate the map updating problem by masking roads that already appeared in the existing map to make sure we only evaluate the quality of the newly discovered portion. For both metrics, F1 is standard, which is the harmonic mean of Precision and Recall, while the definition of Precision and Recall are different.

For GEO metric in the map updating, Precision and Recall are defined as

$$\text{Precision} = \frac{\left|\widetilde{G}_{gt} \cap \widetilde{G}_{\text{inf}}\right|}{\left|\widetilde{G}_{\text{inf}}\right|}, \text{Recall} = \frac{\left|\widetilde{G}_{gt} \cap \widetilde{G}_{\text{inf}}\right|}{\left|\widetilde{G}_{gt}\right|} \tag{7}$$

where $\widetilde{G}_{gt}$ and $\widetilde{G}_{\text{inf}}$ are the set of grid cells where there exist road segments in the ground-truth and inferred completed map but not in the existing map, respectively, i.e., $\widetilde{G}_{gt} = G_{gt}/G_{\text{existing}}$, $\widetilde{G}_{\text{inf}} = G_{\text{inf}}/G_{\text{existing}}$.

For TOPO metric in the map updating, we sample the start grid cells from $\widetilde{G}_{gt} \cap \widetilde{G}_{\text{inf}}$ for N times to make sure the starting road segments are not from the existing map, and calculate the average Precision and Recall

$$\text{Precision} = \frac{\sum_i \frac{\left|G^i_{gt} \cap G^i_{\text{inf}}\right|}{\left|G^i_{\text{inf}}\right|}}{N}, \text{Recall} = \frac{\sum_i \frac{\left|G^i_{gt} \cap G^i_{\text{inf}}\right|}{\left|G^i_{gt}\right|}}{N} \tag{8}$$

where G^i_{gt} and G^i_{inf} are the network-reachable grid cells within 2km in the ground-truth and the inferred completed map when starting from the grid cell sampled at the i-th time.

Note that, when calculating the reachable grid cells, we consider the completed map, since the inferred road segments should not only be reachable to other inferred road segments correctly, but also to the existing road segments as well.

4) Implementation: DelvMap as well as baselines are fully implemented in Python on a docker with 24 Cores@2.5 GHz, 64G memory, and Ubuntu Linux. The baselines from trajectory-driven map updating are re-implemented by us since the source codes are not available. DSFNet as well as other deep learning models are implemented by PyTorch and trained with one NVIDIA Tesla V100 GPU. We have released our code for public use.[4]

Training Details and Hyperparameters: For all deep-learning models, we train 100 epoches and leverage Adam [28]

[4]https://github.com/WangZiyuy/DelvMap

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to perform network training with a learning rate 5e-4 and batch size 8. We also apply a staircase-like schedule by halving the learning rate every 20 epoches. For DelvMap, in the Dual Signal Fusion-based Map Inference step, DSFNet performs the downsampling for four times, and the first ConvBlock of the encoder has 64 filters. λ_1 and λ_2 in (6) are set to 1, and ω is set to 1; in adaptive map completion step, $\mathrm{TH}_{\mathrm{new}}$ and TH_{vm} is set to 32 and 10 m, R is set to 30 m, $S_{\min}$ is set to 1, TH_m is set to 15 m. Hyperparameters of baselines are set as suggested, which are detailed as follows. For CrowdAtlas, the trajectory clustering distance is 20 m, and a soft percentile is 90%. For VitalAlley, the trajectory clustering distance is the same as CrowdAtlas, and the minimum cluster size of the intersection clustering is 5. For GLUE, the size of the rectangle for the direction calculation is 35.96 × 10.79 m, the point clustering distance is 50 m, and the angle threshold is 25°. For HyMU, the clustering distance is 7 m, and the connection distance of the newly inferred road is 42 m. For DeepMG+trFuse, the weights for road centerline prediction loss and road region prediction loss are 0.8 and 0.2, respectively. For RoadTracer+spFuse, the weights for angle loss, action loss, and detection loss are 50, 1, 5, respectively. For SDUNet+Fuse, the hyperparameters are the same as DelvMap.

C. Quantitative Evaluation

1) Overall Performance: The overall performance of DelvMap compared with baselines over all GEO and TOPO metrics as well as the running time is shown in Table II.

From the GEO aspect, trajectory-based map updating algorithms have a higher Precision, while the satellite image-based algorithms have a higher Recall, which met our expectations. Though trajectories are noisy, the existence of them in some regions is still a strong positive and straightforward signal about roads. However, the activities of couriers are related to their delivery tasks, which makes the generated trajectories nearly impossible to cover every roads in the region. Therefore, trajectory-based map updating algorithms lead to a lower Recall compared with the satellite image-based one, which has a global sense of the whole region. In general, the satellite image-based methods are superior in terms of GEO F1. Our method, i.e., DelvMap, combining the advantage of trajectory-based map updating methods (high Precision) as well as the satellite image-based map updating methods (high Recall), achieves the highest F1 score, which outperforms the best baseline by 4.7% in GEO F1.

From TOPO aspect, however, the performance of the satellite image-based methods are deteriorated, which fail the competition with some trajectory-based map updating algorithms due to the lack of actual transition knowledge between roads. We find though DeepMG+trFuse performs better than VitalAlley in GEO F1, it fails the competition for a little bit in TOPO F1. Though DeepMG+trFuse shows better results in GEO F1 quantitatively, we find there are many intermittent roads generated, which does not affect GEO F1 much, but ultimately leads to lower TOPO F1. We guess the reason is that the multitask learning in DeepMG is designed for generating main roads, which is not suitable for generating residential-level roads. DelvMap still outperforms all baselines, since it fuses the inferred map with the existing map based on an adaptive strategy according to the availability of transition trajectories. Specifically, DelvMap outperforms the best baseline by 11.0% in TOPO F1.

From the running time aspects, traditional unsupervised trajectory-based map updating algorithms are not comparable with the deep learning-based approaches, and their practicability in the big trajectory data scenario is of concern. DeepMG+trFuse and DelvMap make trajectory-driven map updating significantly faster by feeding the trajectory source into deep learning models while reserving the transition information for the later fusion module. The running time of DelvMap is 33.6 s, which is faster than most existing map updating methods. Though RoadTracer+spFuse does not require trajectories to fuse maps, the complex inference process of RoadTracer makes it a little bit slower.

2) Ablation Study: The ablation study of DelvMap is also conducted in Table II to validate the effectiveness of the major component of DelvMap. As for the map inference component, after DSFNet is replaced (i.e., DelvMap-nDSFNet), the performance of DelvMap drops the most, which validates the effectiveness of the dual signal fusion model compared with other single-side fusion ones. Note that the finer-grained ablation study of the map inference model would be discussed later. As for the map fusion component, we can observe either removing the spatial proximity-based connection or the trajectory-based connection, the performance would drop. The trajectory-based fusion did make the topology more accurate as the TOPO Precision of DelvMap-trFuse and DelvMap-spFuse showed. DelvMap-trFuse is a little bit slower than DelvMap-spFuse, since the former relies on external trajectories while the latter only depends on the spatial proximity. DelvMap takes a few additional seconds for processing than variants since it would try different strategies to fuse maps to achieve more accurate topology connection.

3) Different Fusion Methods: Since there is no existing multimodality fusion baseline for the map updating, to show the fusion ability of DelvMap, we compare DSFNet with four recent fusion methods and variants of DSFNet for the road extraction quantitatively.

1) Sun et al. [7] is based on the early fusion, which renders the trajectories in the corresponding region as additional channels of the input image tiles, and leverages UNet [13] as the backbone as we did.
2) DeepDualMapper [9] is based on deep fusion, which encodes and decodes the trajectories and satellite images using different UNet [13], and introduces another decoder to fuse the feature maps of two decoders by a gated fusion module.
3) CMMPNet [8] is also based on the deep fusion, which leverages D-LinkNet [29] as the backbone network. It fuses the feature maps at each level of the encoder and the decoder with a message-passing mechanism.
4) DuARE [10] is based on the late fusion, which first individually employs adaptive thinning policies to obtain the roads from trajectories, and a coarse-to-fine convolutional

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TABLE II
OVERALL EVALUATION OF DIFFERENT MAP UPDATING ALGORITHMS

Modality	Method	GEO			TOPO			Inference Time (min)
		Precision	Recall	F1	Precision	Recall	F1	
Trajectory	GLUE [3]	0.6009	0.2199	0.3220	0.7189	0.1345	0.1744	1,789
	CrowdAtlas [2]	0.5847	0.2583	0.3588	0.7055	0.2297	0.2909	2,640
	HyMU [4]	**0.6579**	0.1530	0.2483	0.7508	0.3520	0.4497	3,040
	VitalAlley [5]	0.5978	0.3408	0.4341	0.7160	0.4821	0.5403	4,856
	DeepMG+trFuse [6]	0.6028	0.3801	0.4562	0.6910	0.4843	0.5389	0.59
Satellite Imagery	RoadTracer+spFuse [23]	0.4752	**0.4748**	0.4750	0.6488	0.5171	0.5362	0.62
	SDUNet+spFuse [26]	0.5592	0.4360	0.4900	0.6907	0.4037	0.4545	0.52
Multi-modality	DelvMap-nDSFNet	0.5669	0.4142	0.4786	0.7348	0.3322	0.3930	0.57
	DelvMap-trFuse	0.6245	0.4256	0.5062	0.7636	0.4500	0.5159	0.55
	DelvMap-spFuse	0.6261	0.4347	**0.5131**	0.7383	0.5273	0.5726	**0.50**
	DelvMap (Ours)	0.6261	0.4347	**0.5131**	**0.7673**	**0.5314**	**0.5996**	0.56

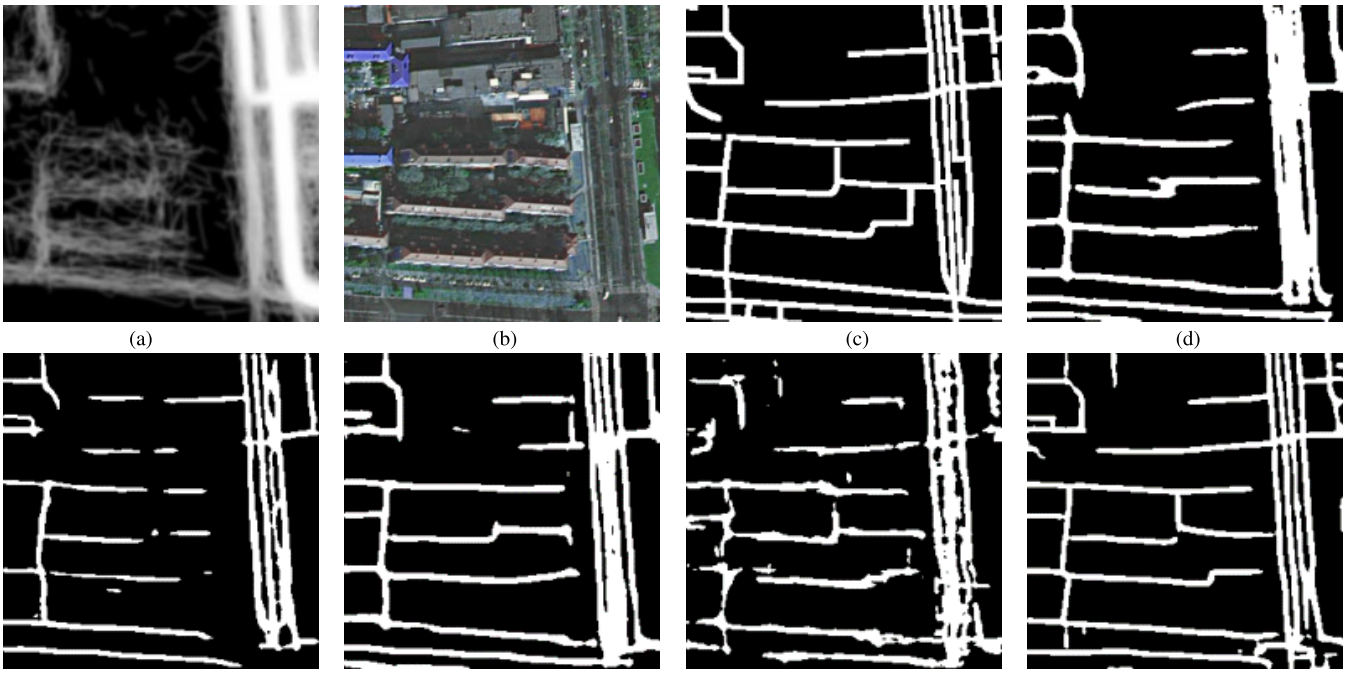


Fig. 6. Comparison of different fusion methods. (a) Trajectories (lines). (b) Satellite image. (c) Ground truth. (d) Sun et al. (early). (e) DeepDualMapper (deep). (f) CMMPNet (deep). (g) DuARE (late). (h) DSFNet (ours).

network to obtain the roads from the satellite images, and then uses a post-processing algorithm to fuse two maps based on the spatial proximity.

The performance of DSFNet and all compared methods is summarized in Table III. We adopt commonly used metrics for the road extraction task, i.e., Precision, Recall, F1, and Intersection over Union (IoU) [8]. Note that those metrics are directly evaluated on the image output of the model, the roads in which might cover several pixels in width.

As observed, the early fusion method does not perform well, since two modalities are shallowly fused. For the deep fusion method, CMMPNet [8] and DeepDualMapper [9] show certain advantages based on the multilevel interaction of the satellite images and trajectory features. They use the convolutional layers to fuse two modalities, which can only capture nearby pixel dependencies. Different from them, STFuse in DSFNet employs attention-based fusion, which captures the pairwise dependencies for all pixels in the feature map. In addition, the BTFuse module in DSFNet makes the model attain negative signals and further boost performance. Although DuARE has the highest Recall, it does not work well in our datasets in terms of F1 and IoU, which is attributed to the unsupervised trajectory-source inference method and the great positioning noises in the residential area.

TABLE III
COMPARISON OF FUSION METHODS

Stage	Method	Pre.	Rec.	F1	IoU
Early	Sun, et al. [7]	0.6410	0.5510	0.5926	0.4212
Deep	CMMPNet [8]	0.6525	0.5666	0.6042	0.4377
	DeepDualMapper [9]	0.6930	0.5628	0.6209	0.4537
Late	DuARE [10]	0.5860	**0.5738**	0.5788	0.4102
/	DSFNet-nSatellite	0.6392	0.3235	0.4296	0.2765
/	DSFNet-nTraj	0.6299	0.5237	0.5719	0.3941
Deep	DSFNet-nBTFuse	0.7175	0.5636	0.6306	0.4639
Hybrid	**DSFNet (Ours)**	**0.7406**	0.5672	**0.6413**	**0.4770**

We also study the effectiveness of each component of DSFNet in Table III. We first study the modality importance by removing the other modality for inference. Comparing DSFNet

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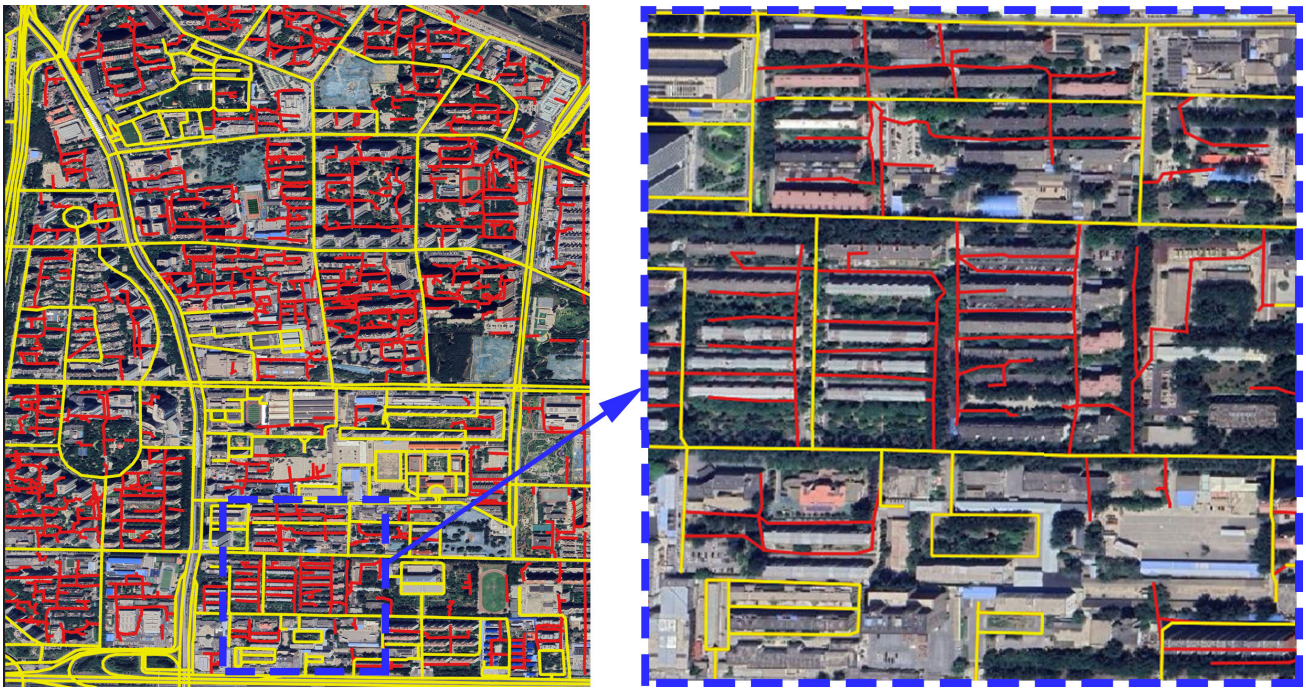


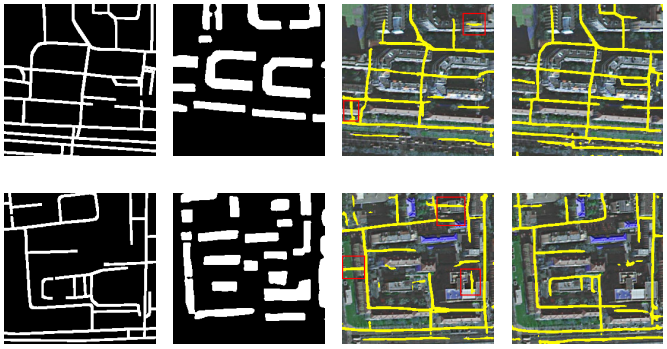
Fig. 7. Completed Maps Produced by DelvMap. Yellow: roads in existing maps, red: inferred roads.

and DSFNet-nSatellite, we can observe Recall drops significantly after removing the satellite image source due to the highly imbalanced data distribution of trajectory. Comparing DSFNet-nTraj and DSFNet-nSatellite, we find the trajectory source is more reliable to infer roads since Precision of DSFNet-nSatellite is higher. We then study fusion modules. Removing STFuse is equivalent to DSFNet-nTraj, which is less effective due to the single modality inference. When BTFuse is removed (DSFNet-nBTFuse), all metrics also drop, which show the advantages of leveraging negative signals.

D. Qualitative Evaluation

1) Different Fusion Methods: We also conduct a qualitative comparison of different multimodal fusion networks in a residential area. The corresponding trajectories (plotted with consecutive lines) and the satellite image are shown in Fig. 6(a) and (b). As can be seen, the early fusion-based method has some difficulties to separate spatially close parallel roads, since trajectories are densely distributed near those roads, and the early fusion strategy can not model such noise accurately. On the contrary, deep fusion-based methods [Fig. 6(e) and (f)] have a better road extraction quality, which have an individual segmentation branch for each modality. However, their Precision and Recall still leave room for improvement. As for DuARE, it generates some spurious roads. We guess the reason is that our trajectories are rather noisy, which brings challenges to the unsupervised trajectory-based map extraction method in DuARE. DSFNet, as shown in Fig. 6(h), also has the best visual results, which is attributed to its dual signal fusion strategy, since it can fully exploit information in the satellite images and trajectories.

2) Completed Maps by DelvMap: The final completed maps in the study region produced by DelvMap is shown in Fig. 7. In the figure, the roads marked by yellow are from existing maps, and the roads marked by red are inferred by DelvMap. We can observe that in the original maps, though the main roads are comprehensively recorded, many residential roads are still missing. With DelvMap, the fine-grained residential roads are inferred and seamlessly connected with main roads.



(a) (b) (c) (d)

Fig. 8. Case studies of the suppression effect of BTFuse. (a) Ground truth. (b) Segmented buildings. (c) DSFNet-nBTFuse. (d) DSFNet.

3) Suppression Effect of BTFuse: In order to understand the suppression effect of BTFuse, we visualize the ground truth, segmented buildings by DSFNet, as well as the road prediction results of DSFNet-nBTFuse and DSFNet of two regions in Fig. 8. As red boxes shown in Fig. 8(c), without BTFuse, some fake roads overlapped with buildings are predicted which could attribute to noisy trajectories. However, the building segmentation branch in DSFNet can successfully recognize those buildings as shown in Fig. 8(b). Leveraging BTFuse module, we find the prediction of roads in corresponding regions is indeed suppressed as shown in Fig. 8(d), which demonstrates the effectiveness of BTFuse.

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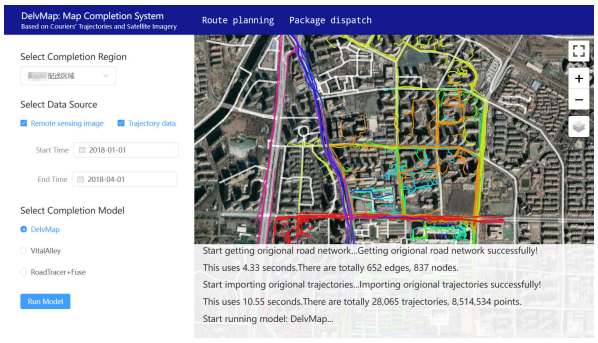


Fig. 9. System deployment.

V. System Deployment

A system based on DelvMap is deployed and used internally in JD Logistics, that can complete the existing map using different methods based on couriers' trajectories and satellite images. In this system, we can directly observe the raw couriers' trajectories and the whole process of different map updating methods. The interface of our system is shown in Fig. 9, which includes three components.

A. Operation View

The left side of the system is the operation view. In this view, the operator can perform multiple operations, which are mainly realized by the following three controls.

1) *Region selection*, which can be used to switch different delivery regions to conduct the map updating.
2) *Data source selection*, that decides which kind of data and how many data would be used to conduct the map updating.
3) *Model selection*, which provides several typical map updating methods including DelvMap for the qualitative comparison. After all options are set, users can click the "Run" button to trigger the map update.

B. Main Map View

The right part is the main map view. Here, we provide a variety of commercial maps, OSM as well as satellite imagery as base maps for users to compare. We can achieve common map operations, such as zooming, layer switching, positioning, etc. During the running of the system, the existing map, the couriers' trajectories, and the completed map will be displayed on different layers.

C. Terminal View

During the execution of the map updating, a view similar to the Terminal will pop up in the lower part of the main map view, which is used to display detailed information during the operation process, such as the running time of each step.

VI. Related Work

A. Trajectory-Based Map Generation and Updating

The map generation has been tackled with or without supervision. There are three main streams of unsupervised approaches according to how the road segments are constructed, which are detailed in [30].

1) Line clustering-based construction [31], [32] uses consecutive points in trajectories to connect road points to form road segments. It suffers from the shortcut issue that should be further handled [32] if the sampling rate is low.
2) KDE-based construction [33], [34] draws the spatial distribution of trajectories into a density map, uses KDE to smooth the value matrix, and extracts road centerlines by skeletoning pixels whose density values are above a certain threshold. Though the shortcut issue can be handled, it is difficult to distinguish spatially close parallel roads with opposite directions.
3) Point clustering-based construction [35], [36] constructs road segments by clustering trajectory points based on spatial proximity and direction similarity. Though this method can handle the low-sampling-rate issue and consider the road directions, it has an essential prerequisite, i.e., the data should be sufficient so that points can be clustered. Though simple and straightforward, unsupervised approaches are difficult to distinguish spatially close parallel roads with the same direction, since the parameters might not be the same for the whole region. To this end, the supervised learning-based method is more appealing. For example, DeepMG [6] uses a UNet-based convolutional network to identify road centerlines.

The existing framework of trajectory-driven map updating has been previously explained in Section III-A. As can be seen, the road segment inference step is similar to the map generation. CrowdAtlas [2] belongs to the line clustering-based method, GLUE [3] belongs to the point clustering-based method, while VitalAlley [5] is a KDE-based method. Different from them, we propose a supervised trajectory-driven map updating method to overcome the GPS positioning issue. DelvMap also incorporates satellite images to overcome issues of imbalanced data distribution and unreliable road indicators by fusing both positive and negative signals.

B. Satellite Image-Based Map Generation and Updating

At present, deep learning models are widely used in various downstream tasks related to remote sensing satellite images, such as classification tasks [37], [38], [39], [40], [41], segmentation tasks [42], [43], [44], change detection [45], [46] and so on. Road detection from satellite images is actively studied in the computer vision community, which is usually modeled as an image semantic segmentation problem and commonly tackled by the convolutional neural network. Since the patch-based segmentation [47], [48] suffers from performance issues and capturing complicated dependencies with neighborhood locations, predicting all pixels at once becomes more popular, e.g., U-Net [13], D-LinkNet [29], DA-RoadNet [49]. Some recent work outputs the map directly from the satellite images, e.g., RoadTracer [25], Sat2Graph [20] and CoANet [50]. For example, CoANet [50] introduces a segmentation branch and a connectivity branch to jointly learn the segmentation and relationship between neighboring pixels to improve the connectivity of road.

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Bastani and Madden [23] extended existing satellite image-based road extraction methods for map updating by introducing a post-processing step based on spatial proximity. P2CNet [51] regards the map updating as a kind of image-based data fusion between the existing maps and the satellite images, which can only guarantee the geometrical accuracy of the completed map, and semantic attributes of road in the existing maps might be lost. Different from them, we propose to leverage trajectories to improve geometrical accuracy of the road extraction and the topological accuracy in regions where the trajectories are sufficient.

C. Multimodal Fusion-Based Map Generation

Recently, the multimodal fusion-based map generation is becoming more popular, since the information from multiple views can be leveraged to achieve a higher recall and precision. Multimodal fusion methods can be divided into three categories, as used in the general MDL framework introduced by [52].

1) Early-fusion, which renders the other data source as additional image channels and concatenates them with the satellite images as the input of image segmentation network [7].
2) Deep-fusion, which designs the information fusion module to fuse/exchange information in intermediate feature maps from each individual encoding network of each data source [8], [9], [41], [53].
3) Late-fusion, which fuses the information after the main branches of different subnetworks or after the network inference.

Hong et al. [24] proposes to fuse different types of features extracted from hyperspectral images by GCN and CNN, for the land use classification, and [10] first individually detects roads from each data source based on different methods, and then leverages a post-processing algorithm to merge generated maps into one. Different from them, we are the first work to introduce the multimodal fusion into the map updating problem. In addition, we propose a novel hybrid fusion strategy, which not only fuses positive signals (i.e., roads) for the extraction as existing work did, but also exploits the negative signals (i.e., buildings) in the satellite images.

VII. Conclusion

In this article, we study the problem of map updating based on couriers' trajectories and satellite images. We propose a novel two-step framework DelvMap. The first step, i.e., Dual Signal Fusion-based Map Inference, leverages two modalities to generate an inferred map via a DSFNet, which fuses both the positive and the negative signals in satellite images with trajectories. The second step, i.e., adaptive map completion, connects newly discovered road segments with the existing map with different merging strategies, which is adaptive to the number of traversing trajectories. Experiments show DelvMap outperforms the best baseline by 11.0% in TOPO F1. A system is also implemented to demonstrate the real-world applications of DelvMap.

We anticipate possible future research directions from following three perspectives.

1) From the problem formulation aspect, we only focus on discovering missing or newly constructed roads and connecting them to the existing map similar to most existing map updating work. It is also possible to find deconstructed or incorrect roads in the existing maps when the temporal sliding window is applied to the data, which remains a future work if more data over a long time period is available.
2) From the methodology aspect, DelvMap employs a two-step strategy involving road extraction and map completion. Solving the map completion problem by one holistic model might further boost the performance given its advantage of the end-to-end optimization. Besides, DelvMap only considers buildings as negative signals, while incorporating the detection of other geospatial entities that could be traversed by humans, e.g., the lawn, may also improve the performance of DSFNet if the lawn labels are sufficient.
3) For the data aspect, with the advances of different sensing technologies, fusing a wide range of modality data to complete the residential maps can be envisioned.

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References

[1] H. Wen et al., "Enough waiting for the couriers: Learning to estimate package pick-up arrival time from couriers' spatial–temporal behaviors," *ACM Trans. Intell. Syst. Technol.*, vol. 14, no. 3, pp. 1–22, Jun. 2023.
[2] Y. Wang, X. Liu, H. Wei, G. Forman, and Y. Zhu, "CrowdAtlas: Self-updating maps for cloud and personal use," in *Proc. 11th Annu. Int. Conf. Mobile Syst., Appl., services*, Jun. 2013, pp. 27–40.
[3] H. Wu, C. Tu, W. Sun, B. Zheng, H. Su, and W. Wang, "GLUE: A parameter-tuning-free map updating system," in *Proc. 24th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2015, pp. 683–692.
[4] T. Wang, J. Mao, and C. Jin, "HyMU: A hybrid map updating framework," in *Proc. Int. Conf. Database Syst. Adv. Appl.* Cham, Switzerland: Springer, 2017, pp. 19–33.
[5] C. Cao, Z. Liu, M. Li, W. Wang, and Z. Qin, "Walkway discovery from large scale crowdsensing," in *Proc. 17th ACM/IEEE Int. Conf. Inf. Process. Sensor Netw. (IPSN)*, Apr. 2018, pp. 13–24.
[6] S. Ruan et al., "Learning to generate maps from trajectories," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 1, 2020, pp. 890–897.
[7] T. Sun, Z. Di, P. Che, C. Liu, and Y. Wang, "Leveraging crowdsourced GPS data for road extraction from aerial imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7509–7518.
[8] L. Liu, Z. Yang, G. Li, K. Wang, T. Chen, and L. Lin, "Aerial images meet crowdsourced trajectories: A new approach to robust road extraction," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3308-3322, Jul. 2023.
[9] H. Wu, H. Zhang, X. Zhang, W. Sun, B. Zheng, and Y. Jiang, "DeepDualMapper: A gated fusion network for automatic map extraction using aerial images and trajectories," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 1, 2020, pp. 1037–1045.
[10] J. Yang et al., "DuARE: Automatic road extraction with aerial images and trajectory data at Baidu maps," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2022, pp. 4321–4331.
[11] P. Newson and J. Krumm, "Hidden Markov map matching through noise and sparseness," in *Proc. 17th ACM SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, 2009, pp. 336–343.
[12] S. Ruan et al., "Doing in one go: Delivery time inference based on couriers' trajectories," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 2813–2821.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:23:29 UTC from IEEE Xplore. Restrictions apply.

[13] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[14] H. Xu and K. Saenko, "Ask, attend and answer: Exploring question-guided spatial attention for visual question answering," in *Proc. 14th Eur. Conf. Comput. Vision*, Amsterdam, The Netherlands, Springer, 2016, pp. 451–466. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-46478-7_28#citeas

[15] Z. Yu, J. Yu, Y. Cui, D. Tao, and Q. Tian, "Deep modular co-attention networks for visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6281–6290.

[16] J. Lu, J. Yang, D. Batra, and D. Parikh, "Hierarchical question-image co-attention for visual question answering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2016/hash/9dcb88e0137649590b755372b040afad-Abstract.html

[17] D.-K. Nguyen and T. Okatani, "Improved fusion of visual and language representations by dense symmetric co-attention for visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6087–6096.

[18] Q. Wu, P. Wang, C. Shen, I. Reid, and A. V. D. Hengel, "Are you talking to me? Reasoned visual dialog generation through adversarial learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6106–6115.

[19] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[20] S. He et al., "Sat2Graph: Road graph extraction through graph-tensor encoding," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 51–67.

[21] T. Y. Zhang and C. Y. Suen, "A fast parallel algorithm for thinning digital patterns," *Commun. ACM*, vol. 27, no. 3, pp. 236–239, Mar. 1984.

[22] W. Shi, S. Shen, and Y. Liu, "Automatic generation of road network map from massive GPS, vehicle trajectories," in *Proc. 12th Int. IEEE Conf. Intell. Transp. Syst.*, Oct. 2009, pp. 1–6.

[23] F. Bastani and S. Madden, "Beyond road extraction: A dataset for map update using aerial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 11885–11894.

[24] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.

[25] F. Bastani et al., "RoadTracer: Automatic extraction of road networks from aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4720–4728.

[26] M. Yang, Y. Yuan, and G. Liu, "SDUNet: Road extraction via spatial enhanced and densely connected UNet," *Pattern Recognit.*, vol. 126, Jun. 2022, Art. no. 108549.

[27] J. Biagioni and J. Eriksson, "Inferring road maps from global positioning system traces: Survey and comparative evaluation," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2291, no. 1, pp. 61–71, Jan. 2012.

[28] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[29] L. Zhou, C. Zhang, and M. Wu, "D-LinkNet: LinkNet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 182–186.

[30] S. Ruan, K. Xiong, S. Wang, J. Geng, J. Bao, and Y. Zheng, "A survey of urban geographic information inference driven by crowd-sourced spatio-temporal data," *Acta Electronica Sinica*, vol. 51, no. 8, pp. 2238–2259, 2023.

[31] L. Cao and J. Krumm, "From GPS traces to a routable road map," in *Proc. 17th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.* New York, NY, USA: ACM, Nov. 2009, pp. 3–12.

[32] R. Stanojevic, S. Abbar, S. Thirumuruganathan, S. Chawla, F. Filali, and A. Aleimat, "Robust road map inference through network alignment of trajectories," in *Proc. ICDM*. Philadelphia, PA, USA: SIAM, 2018, pp. 135–143.

[33] J. Biagioni and J. Eriksson, "Map inference in the face of noise and disparity," in *Proc. 20th Int. Conf. Adv. Geographic Inf. Syst.* New York, NY, USA: ACM, Nov. 2012, pp. 79–88.

[34] X. Yang, L. Tang, C. Ren, Y. Chen, Z. Xie, and Q. Li, "Pedestrian network generation based on crowdsourced tracking data," *Int. J. Geograph. Inf. Sci.*, vol. 34, no. 5, pp. 1051–1074, May 2020.

[35] C. Chen, C. Lu, Q. Huang, Q. Yang, D. Gunopulos, and L. Guibas, "City-scale map creation and updating using GPS collections," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 1465–1474.

[36] Z. Fang, G. Wang, X. Xie, F. Zhang, and D. Zhang, "Urban map inference by pervasive vehicular sensing systems with complementary mobility," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 5, no. 1, pp. 1–24, Mar. 2021.

[37] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[38] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[39] X. Sun, P. Wang, C. Wang, Y. Liu, and K. Fu, "PBNet—Part-based convolutional neural network for complex composite object detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 50–65, Mar. 2021.

[40] W. Li, H. Chen, Q. Liu, H. Liu, Y. Wang, and G. Gui, "Attention mechanism and depthwise separable convolution aided 3DCNN for hyperspectral remote sensing image classification," *Remote Sens.*, vol. 14, no. 9, p. 2215, May 2022.

[41] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[42] M. D. Hossain and D. Chen, "Segmentation for object-based image analysis (OBIA): A review of algorithms and challenges from remote sensing perspective," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 115–134, Apr. 2019.

[43] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[44] H. Wang, X. Chen, T. Zhang, Z. Xu, and J. Li, "CCTNet: Coupled CNN and transformer network for crop segmentation of remote sensing images," *Remote Sens.*, vol. 14, no. 9, p. 1956, Apr. 2022.

[45] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[46] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5607514. [Online]. Available: https://ieeexplore.ieee.org/document/9491802

[47] V. Mnih and G. Hinton, "Learning to label aerial images from noisy data," in *Proc. 29th Int. Conf. Mach. Learn. (ICML)*, vol. 1, 2012, pp. 567–574.

[48] V. Mnih, "Machine learning for aerial image labeling," Ph.D. dissertation, Canada, 2013. [Online]. Available: https://dl.acm.org/doi/10.5555/2604667

[49] J. Wan, Z. Xie, Y. Xu, S. Chen, and Q. Qiu, "DA-RoadNet: A dual-attention network for road extraction from high resolution satellite imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6302–6315, 2021.

[50] J. Mei, R.-J. Li, W. Gao, and M.-M. Cheng, "CoANet: Connectivity attention network for road extraction from satellite imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 8540–8552, 2021.

[51] J. Mei, R.-J. Li, W. Gao, and M.-M. Cheng, "CoANet: Connectivity attention network for road extraction from satellite imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 8540–8552, 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9563125

[52] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020.

[53] J. Zhang, Q. Hu, J. Li, and M. Ai, "Learning from GPS trajectories of floating car for CNN-based urban road extraction with high-resolution satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1836–1847, Mar. 2021.



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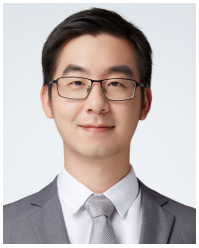
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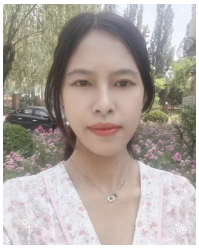


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